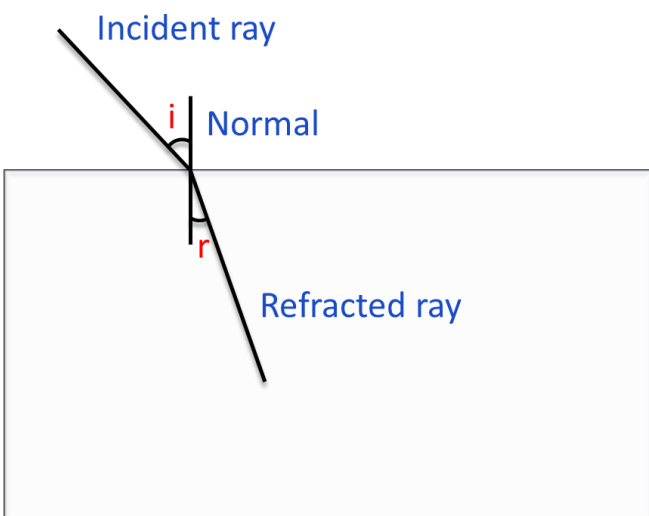
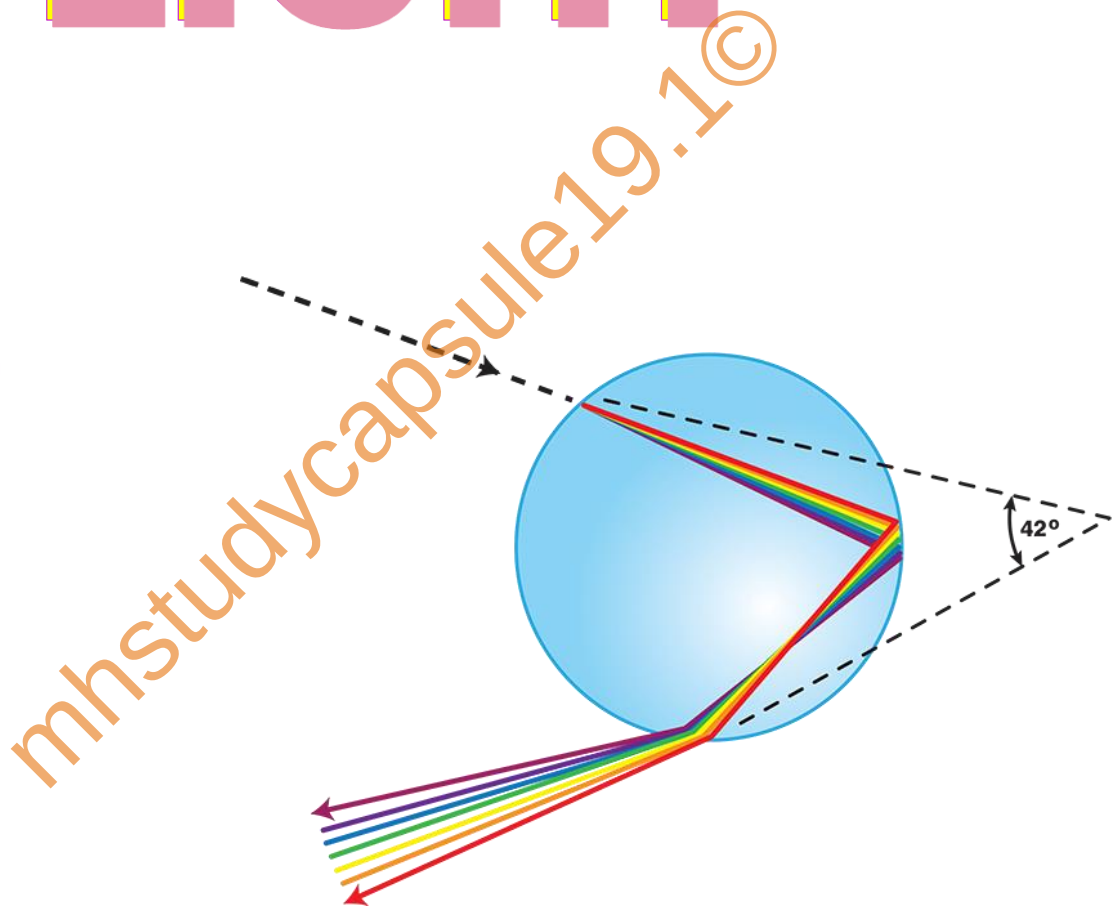


Chapter 6

Refraction of

LIGHT



1. Fill in the blanks and EXPLAIN the completed sentences.

- Refractive index depends on the velocity of light.
- The change in direction of light rays while going from one medium to another is called refraction.

EXPLANATION

- The refractive index of a material is a measure of how much light slows down when passing through it, hence it is directly dependent on the velocity of light in that medium.
- Refraction is when light bends as it goes from one medium to another, like from air to water. This happens because light travels at different speeds in different mediums. The change in speed makes the light change direction.

2. Prove the following statements.

- If the angle of incidence and angle of emergence of a light ray falling on a glass slab are 'i' and 'e' respectively, prove that $i = e$.

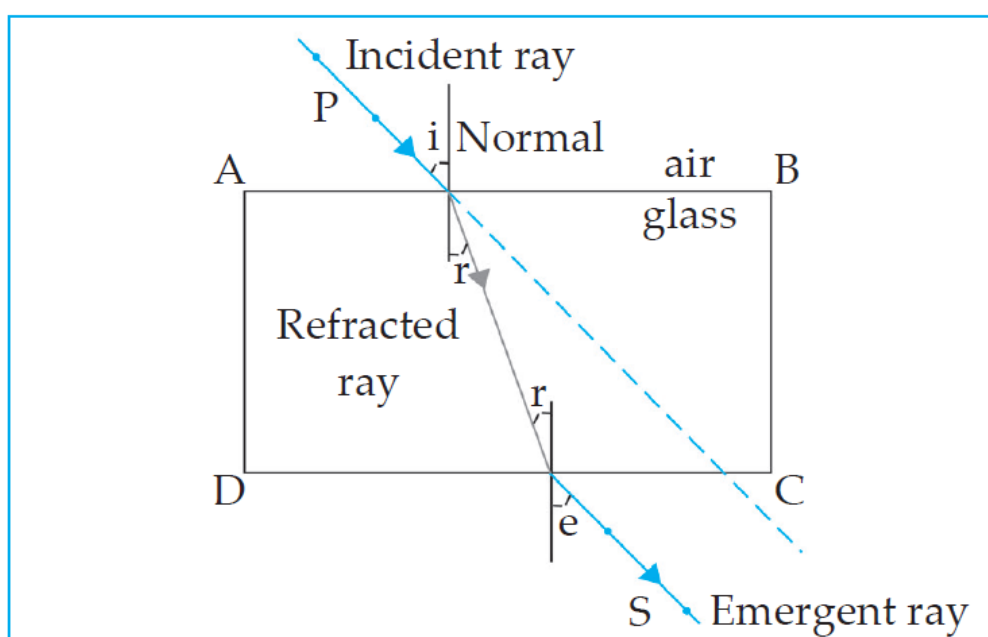


Diagram to be drawn in exam.

Refractive index of second medium w.r.t first medium - 1n_2

Refractive index of first medium w.r.t second medium - 2n_1

$${}^1n_2 = \frac{1}{{}^2n_1}$$

Let 1st medium be air and 2nd medium be glass.

$${}^a n_g = \frac{1}{{}^g n_a} \quad \text{.....(i)}$$

At surface AB

$${}^a n_g = \frac{\sin i}{\sin r} \quad \text{.....(ii)}$$

At surface CD

$${}^g n_a = \frac{\sin r}{\sin e} \quad \text{.....(iii)}$$

Substituting (ii) and (iii) in (i)

$$\left(\frac{\sin i}{\sin r} \right) = \frac{1}{\left(\frac{\sin r}{\sin e} \right)}$$

$$\frac{\sin i}{\sin r} = \frac{\sin e}{\sin r}$$

$$\sin i = \sin e$$

$$i = e$$

b. A rainbow is the combined effect of the refraction, dispersion, and total internal reflection of light

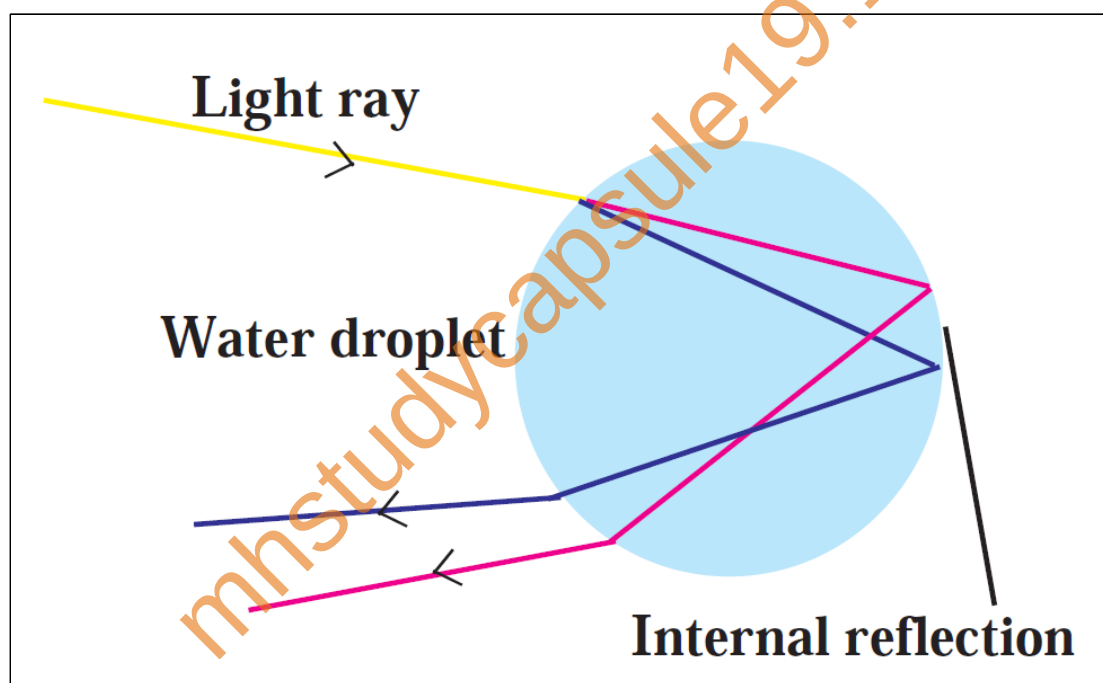


Diagram to be drawn in exam.

- i. Rainbow is a beautiful natural phenomenon. It is the combined effect of a number of natural processes.
- ii. It is the combined effect of dispersion, refraction and total internal reflection of light. It can be seen mainly after a rainfall.
- iii. Small droplets of water act as small prisms. When light rays from the Sun enter these droplets, it gets refracted and dispersed.

- iv. Then there is internal reflection as shown in the diagram, and after that once again the light gets refracted while coming out of the droplet.
- v. All these three processes together produce the rainbow.

3. Mark the correct answer in the following questions.

1. What is the reason for the twinkling of stars?
 - a. Explosions occurring in stars from time to time
 - b. Absorption of light in the earth's atmosphere
 - c. Motion of stars
 - d. **Changing refractive index of the atmospheric gases**
2. We can see the Sun even when it is little below the horizon because of
 - a. Reflection of light
 - b. **Refraction of light**
 - c. Dispersion of light
 - d. Absorption of light
3. If the refractive index of glass with respect to air is $\frac{3}{2}$, what is the refractive index of air with respect to glass?

is?

a. $\frac{1}{2}$

b. 3

c. $\frac{1}{3}$

d. $\frac{2}{3}$

4. Solve the following examples.

- a. If the speed of light in a medium is 1.5×10^8 m/s, what is the absolute refractive index of the medium?

Given:

$$V_m = 1.5 \times 10^8 \text{ m/s}$$

$$V_{\text{vac}} = 3 \times 10^8 \text{ m/s}$$

To find:

Absolute refractive index of the medium, ${}^{\text{vac}}n_m = ?$

Formula:

$${}^{\text{vac}}n_m = \frac{C}{V_m}$$

Solution:

$${}^{\text{vac}}n_m = \frac{3 \times 10^8}{1.5 \times 10^8}$$

$${}^{\text{vac}}n_m = 2$$

Absolute refractive index of the medium is 2.

b. If the absolute refractive indices of glass and water are $\frac{3}{2}$ and $\frac{4}{3}$ respectively, what is the refractive index of glass with respect to water?

Given:

$$n_g = \frac{3}{2} = \frac{C}{V_g}$$

$$n_w = \frac{4}{3} = \frac{C}{V_w}$$

To find:

$${}^w n_g = ?$$

Formula:

$${}^w n_g = \frac{V_w}{V_g}$$

Solution:

$${}^w n_g = \frac{V_w}{V_g}$$

Multiplying and dividing with 'C'

$${}^w n_g = \frac{V_w}{V_g} \times \frac{C}{C}$$

$${}^w n_g = \frac{C}{V_g} \times \frac{V_w}{C}$$

$${}^w n_g = \frac{3}{2} \times \frac{3}{4} \quad [\text{given}]$$

$${}^w n_g = \frac{9}{8}$$

Refractive index of glass w.r.t water
is $\frac{9}{8}$.

INTEXT QUESTIONS

1. What is meant by reflection of light?

When rays of light fall on a smooth surface they turn back. This is called reflection of light.

2. What are the laws of reflection?

- i. The incident ray, reflected ray and the normal are all in the same plane.
- ii. The incident ray and reflected ray are on opposite sides of the normal.
- iii. The angle of incidence and angle of reflection are of equal measure.

3. Will light travel through a glass slab with the same velocity as it travels in air?

- i. Refractive index depends on the velocity of light.
- ii. As refractive index increases, velocity of light decreases.
- iii. Since refractive index of glass is greater than air, the velocity of light in glass is lesser than that in air.

4. Will the velocity of light be same in all media?

- i. Refractive index depends on the velocity of light.
- ii. Refractive index is different for different media.
- iii. Velocity of light will also be different for different media.

5. If the refractive index of second medium with respect to first medium is 1n_2 and that of third medium with respect to second medium is 2n_3 , what and how much is 1n_3 ?

Refractive index of 2nd medium w.r.t. 1st

$${}^1n_2 = \frac{V_1}{V_2} \quad \text{.....(i)}$$

Refractive index of 3rd medium w.r.t. 2nd

$${}^2n_3 = \frac{V_2}{V_3} \quad \text{.....(ii)}$$

Similarly, refractive index of 3rd medium w.r.t. 1st

$${}^1n_3 = \frac{V_1}{V_3} \quad \text{.....(iii)}$$

Multiplying and dividing equation (iii) with V_2

$${}^1n_3 = \frac{V_1}{V_3} \times \frac{V_2}{V_2}$$

$${}^1n_3 = \frac{V_2}{V_3} \times \frac{V_1}{V_2}$$

From (i) and (ii)

$${}^1n_3 = {}^2n_3 \times {}^1n_2$$

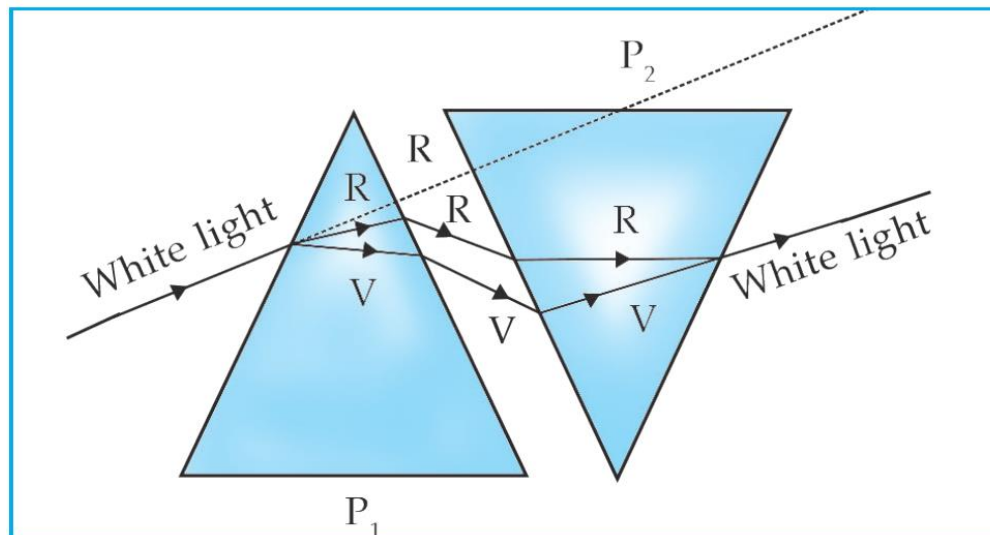
6. Have you seen a mirage which is an illusion of the appearance of water on a hot road or in a desert?

- i. Yes. I have seen mirage on hot road during summer days.
- ii. The air near the hot road or desert surface is hot and hence rarer than the air above.
- iii. As height increases, the refractive index also increases.
- iv. The direction of light rays keeps changing due to refraction and appears to be coming from the image of the object on the ground.
- v. This is called a mirage.

7. Have you seen that objects beyond and above a holi fire appear to be shaking? Why does this happen?

- i. Yes, this happens due to a phenomenon called "hot air shimmer" or "thermal refraction."
- ii. The intense heat from the fire causes the air above it to become much hotter than the surrounding air, creating regions of varying refractive index.
- iii. Light passing through these regions experiences rapid changes in direction as it moves from one layer of hot air to another, causing the appearance of shaking or distortion when looking at objects beyond the fire.

8. From incident white light how will you obtain white emergent light by making use of two prisms?



- i. Take a glass prism, Allow white light to fall on it. Obtain a spectrum.
- ii. Take a second identical glass prism. Place it parallel to the first prism in an upside-down position with the first prism.
- iii. Allow the colours of the spectrum to pass through the second prism.
- iv. It is found then, that a beam of white light emerges from the other side of the second prism.

9. You must have seen chandeliers having glass prism. The light from a tungsten bulb gets dispersed while passing through these prisms and we see coloured spectrum. If we use on LED light instead of tungsten bulb, will we be able to see the same effect?



Image just for reference.

- i. Yes, you can see a similar effect with LED lights and glass prisms as you would with tungsten bulbs.
- ii. Both tungsten bulbs and LED lights emit white light, which is a combination of various colors.
- iii. When this white light passes through the glass prisms of the chandelier, it gets dispersed into its component colors, creating a beautiful colored spectrum.

Solved Examples

1. The absolute refractive index of water is 1.36. What is the velocity of light in water? (velocity of light in vacuum = $C = 3 \times 10^8$ m/s)

Given:

$$V_{\text{vac}} = 3 \times 10^8 \text{ m/s}$$

$$n_w = 1.36$$

To find:

Velocity of light in water, $V_w = ?$

Formula:

$${}^{\text{vac}}n_w = \frac{C}{V_w}$$

Solution:

$$1.36 = \frac{3 \times 10^8}{V_w}$$

$$V_w = \frac{3 \times 10^8}{1.36}$$

$$V_w = 2.2058 \times 10^8 \text{ m/s}$$

$$V_w \approx 2.1 \times 10^8 \text{ m/s}$$

2. Light travels with a velocity 1.5×10^8 m/s in a medium. On entering second medium its velocity becomes 0.75×10^8 m/s. What is the refractive index of the second medium with respect to the first medium?

Given:

$$V_1 = 1.5 \times 10^8 \text{ m/s}$$

$$V_2 = 0.75 \times 10^8 \text{ m/s}$$

To find:

Refractive index of 2nd medium w.r.t 1st medium, ${}^1n_2 = ?$

Formula:

$${}^1n_2 = \frac{V_1}{V_2}$$

Solution:

$${}^1n_2 = \frac{1.5 \times 10^8}{0.75 \times 10^8}$$

$${}^1n_2 = 2$$

Refractive index of 2nd medium w.r.t 1st medium is 2



Previous Year Questions

1. If the speed of light in a medium is 1.5×10^8 m/s, what is the absolute refractive index of the medium?

Exercise Question 4 – a

2. Stars twinkle but we do not see the twinkling of planets.

- Stars appear to be point sources because they are at a very large distance from us.
- Planets are much closer to us as compared to stars. They do not appear as point sources but appear as a collection of point sources.
- Because of changes in atmospheric refractive index, the position as well as the brightness of individual point sources change, but the average position and total average brightness remains unchanged.
- Thus, stars twinkle, but planets do not twinkle.

3. Star appears to be twinkling at night. Give Reason

- Stars are self luminous and can be seen at night in the absence of sunlight.
- They appear to be point sources as they are at a very large distance from us.
- Because of the motion of atmospheric air and changing air density and temperature, the refractive index of air keeps changing continuously.
- Because of this, the position and brightness of the stars keeps changing and so the stars appear to be twinkling.

4. Explain the laws of refraction.

The laws of refraction are as follows:

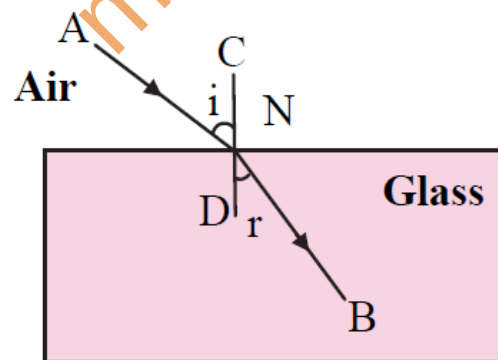
- i. The incident ray and the refracted ray are on the opposite sides of the normal at the point of incidence and all three lie in the same plane.
- ii. For a given pair of media, the ratio of the sine of the angle of incidence to the sine of the angle of refraction is constant.

If 'i' is the angle of incidence and 'r' is the angle of refraction, then $\frac{\sin i}{\sin r} = \text{constant} = n$

n is called the refractive index of the second medium with respect to the first medium. This second law is also called Snell's law.

5. Observe the given figure and answer the following questions:

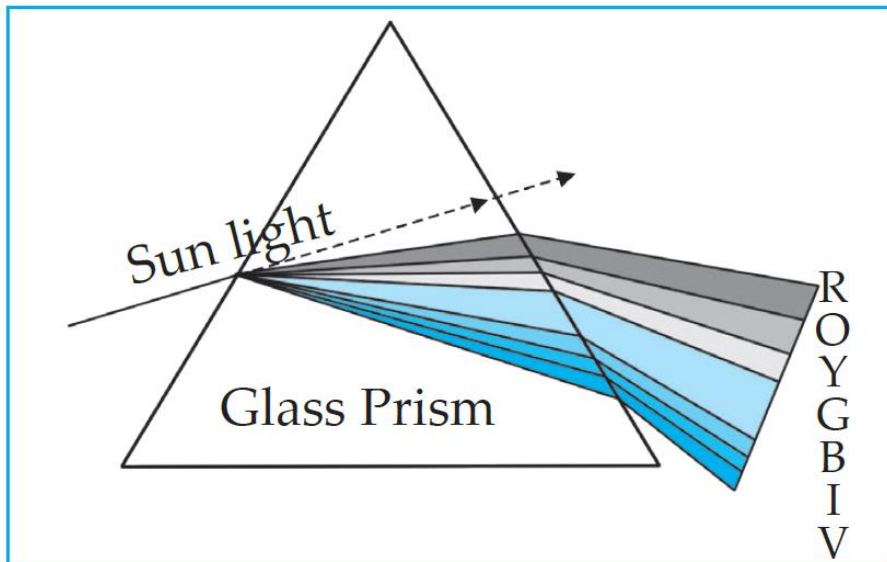
- a. Name the process represented by the figure.
- b. State the two laws related to the process.



- a. The process represented in the figure is refraction of light.
- b. Refer the previous answer.

6. a. Draw a neat labelled diagram of 'dispersion of white light through glass prism'.
- b. Which coloured ray is the least deviated?
- c. Which coloured ray is the most deviated?

a.



- b. Red coloured ray is least deviated.
- c. Violet coloured ray is most deviated.

7. With the help of a neat labelled diagram prove that a rainbow is the combined effect of the refraction, dispersion, and total internal reflection of light.

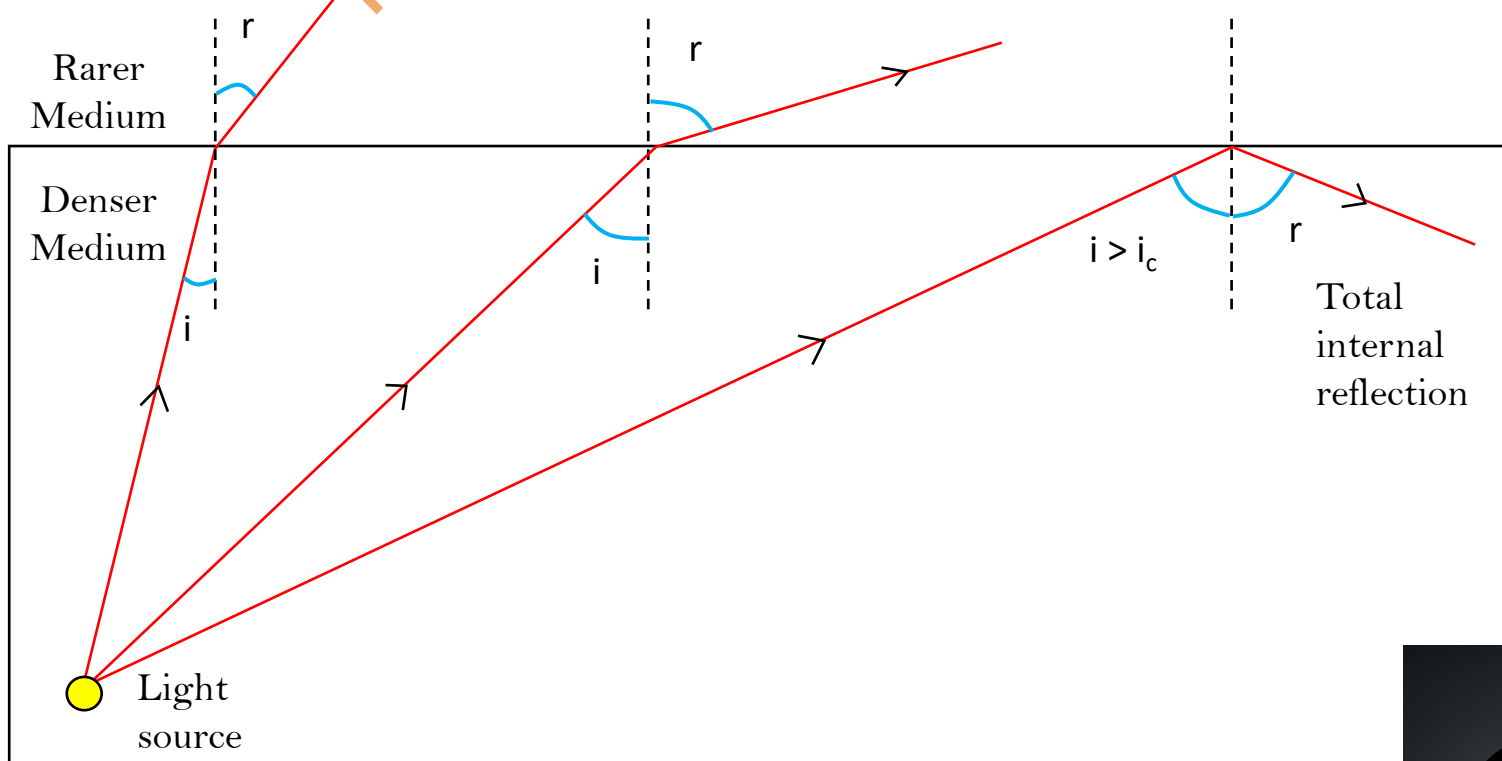
Exercise Question 2 -b

8. The observation made by Swarali while doing the experiment are given below. Based on these write answers to the questions: Swarali found that the light ray travelling from the denser medium to rarer medium goes away from the normal. If the angle of incidence (i) is raised by Swarali, the angle of refraction (r) went on increasing. However, after certain value of the angle of incidence the light ray is seen to return into the denser medium.

- i. **What is the specific value of i called?**
- ii. **What is the process of reflection of incident ray into denser medium called?**
- iii. **Draw the diagrams of three observations made by Swarali.**

- i. The specific value of ' i ' is called Critical angle.
- ii. The process of reflection of incident ray into denser medium is called Total internal reflection.

iii.



9. Observe the figure and name the ray AB, ray CD, ray GH:

- i. Ray AB – Incident ray
- ii. Ray CD – Refracted ray
- iii. Ray GH – Emergent ray

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